Load Balancing :-

In computing load balancing refers to the process of distributing the tasks over a set of resources.

Load balancing can optimize the response time for each task.

⇒ Two approaches to load balancing exist:

- Static algorithms
- Dynamic algorithms.

Goals of load balancing

- Achieve optimal resource utilization
- Maximize throughput
- Minimize response time
- Avoid overload
- Avoid crashing

Types of Loadbalancing

⇒ There are two types of load balancing.

- Static load balancing
- Dynamic load balancing

Static load balancing:

⇒ It is the type of load Balancing which is often referred to as the mapping problem, the task is to map a static process graph into a fixed hardware topology in order to minimise dilation and process load differences.

⇒ In static load balancing, the distribution of tasks among processing units is predetermined or fixed before the execution of the program.

Implementation:

⇒ Task assignment is typically based on assumptions about the workload or characteristics of the tasks.

⇒ This can be done manually or using heuristics during program design.

Pros:

- Simplicity
- Suitable when the workload is known in advance.

Cons:

- If the workload varies significantly during execution, some processing units may be underutilized while other are overloaded.

Dynamic load balancing

⇒ In dynamic load balancing, the distribution of tasks adjusted during the execution of the program based on the current state of the system.

Pros:

- Ensures utilization of resources
- Can handle varying workloads
- Promote better performance

Implementation:

- Algorithms continuously monitor the systems workload and redistribute tasks dynamically.

Cons:

Complexity

Overhead

ns on load balancing

Q. Write purpose of load balancing in distributed system?

- Security
- protect applications from emerging threats
- SSL offload
- Traffic Compression.

. Name load balancing approaches Distributed systems.

- Round Robin
- Least Connections
- Least Time
- Hash
- IP hash.

Q. write advantages of load balancing?

⇒ Load balancers minimize server response time and maximize throughput.

⇒ Load balancer ensures high availability and reliability by sending requests only to auline servers.

⇒ Load balancers do Continuous health checks to monitor the server Capability of handling the request.

Q. How to achieve synchronization in distributed systems?

Through clocks.

Physical clocks used to adjust the time of nodes.

Each node in the system share its time with other node.

CamScanner

Q. Name the issues related to load balancing in distributed system?

⇒ Performance degradation
⇒ Job selection
⇒ Load Level Comparison
⇒ Load estimation
⇒ Performance indices
⇒ Availability
⇒ Stability
⇒ Security
⇒ Homogeneous nodes

Q. List sheduling techniques in load balancing?
Distribute System

The techniques that are used for sheduling the process in distributed systems are:

- Task Assignment Approach
- Load balancing Approach
- Load Sharing Approach

CS CamScanner